Mr Dimelow

3 Processes, techniques and specialist tools

3.4 Permanent and Semi-Permanent Joining Techniques

a) adhesives – contact adhesive, acrylic cement, epoxy resin, polyvinyl acetate (PVA), hot melt glue, cyanoacrylate (superglue), polystyrene cement (including use of specialist tools)

Definition: Adhesives

An ***adhesive*** is a substance—such as *glue, cement, paste, or mortar*—used to join ***<u>two or more</u>*** surfaces together by resisting separation. They work by ***<u>bonding</u>*** materials through chemical, dispersive, or electrostatic forces, commonly used for sticking objects together, in manufacturing (e.g., aerospace, construction), or as sticky coatings.

Adhesives can be natural (animal glue, starch) or synthetic (polyurethane, acrylic). They are classified as reactive (e.g., epoxies) or non-reactive (e.g., drying adhesives) based on how they harden.

Type: Contact Adhesive

Contact adhesive (or contact cement) is a *high-strength, flexible glue* used for bonding materials like *wood, metal, plastic, and rubber*, instantly upon contact. *Applied to both surfaces and dried, it requires no clamping, making it ideal for laminates, veneers, and repairs.* Popular types include Evo-Stik, Gorilla, and spray-on versions.

Video: How to use Contact adhesive

https://www.youtube.com/watch?v=mLhgAPIY7Iw

Type: Contact Adhesive

Key Features and Usage:

- **Application Method:** *Apply to both surfaces, wait for it to become touch-dry (usually 10-20 minutes), then press together firmly.*
- **Instant Bond:** *Creates an immediate, permanent bond, so accurate positioning is essential.*
- **Versatility:** *Ideal for bonding laminates, rubber, leather, cork, and metals, often used for countertop edging or large surface bonds.*
- **Durability:** *Provides strong, flexible, and often waterproof or heat-resistant bonds.*
- **Forms:** *Available in tubes (for small repairs), tins (for brushing/spreading), and sprays.*

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Type: Contact Adhesive

Popular Brands & Products:

- **EVO-STIK Impact Adhesive:** *A versatile, high-strength adhesive.*
- **Gorilla Contact Adhesive:** *Known for being waterproof and drying clear.*
- **Thixofix:** *A popular non-drip formula.*
- **Bostik All Purpose:** *Fast-setting for general repairs.*











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Type: Contact Adhesive

Tips for Success:

- **Clean Surfaces:** *Ensure materials are free of dust, grease, and moisture before application.*
- **Temperature:** *Apply in a well-ventilated area at room temperature for best results.*
- **Clean Up:** *Use a dedicated contact adhesive cleaner to remove excess, as it dries quickly.*

Type: Acrylic Cement

Acrylic cement glue is a ***solvent-based adhesive*** that *welds acrylic (Perspex/Plexiglas)* by melting the surfaces together to create a strong, clear, and permanent bond. Primarily applied via capillary action for edge-gluing, it sets in minutes but takes 24–48 hours to fully cure. Popular types include water-thin solvents (Weld-On 3/4) and thicker, gap-filling cements (Weld-On 16).

Solvent Cement (Thin)



Tensol 12 - Body Cement (Thick)



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Type: Acrylic Cement

Video: How to use Solvent Acrylic Glue

https://www.youtube.com/watch?v=M4K8yo-_hSc

Video: How to use Tensol 12 Acrylic Glue

https://www.youtube.com/watch?v=JRtS35jxw9A

Solvent Cement (Thin)



Tensol 12 - Body Cement (Thick)



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Type: Acrylic Cement

Key Types and Usage

- **Solvent Cement (Thin):** *Water-thin liquids (e.g., Liquid Solvent Cement, Weld-On 3 or 4) are applied to joints using a needle applicator, using capillary action to flow into tight, well-fitted joints.*
- **Body Cement (Thick):** *Thicker, syrupy cements (e.g., Tensol No 12, Weld-On 16, Acrifix 192) are used for filling small gaps or for vertical surfaces, as they do not run as easily.*
- **Two-Part Cement:** *For industrial or high-strength, weather-resistant applications, such as Anglosol 700.*

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Type: Acrylic Cement

Application Process

1. **Preparation:** *Ensure surfaces are clean, dry, and flat. The cement will not fill large gaps, requiring a precise fit.*
2. **Application:** *Use a needle applicator (e.g., Hypo 25 or 65) to apply a thin bead of cement along the joint.*
3. **Bonding:** *The solvent softens the plastic instantly. Assemble pieces and apply slight pressure for 1–2 minutes.*
4. **Curing:** *While it sets in minutes, full strength is achieved in 24–48 hours.*

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Type: Acrylic Cement

Safety and Tips

- **Safety:** *Always use in a well-ventilated area, preferably outdoors, and wear protective gloves and safety glasses.*
- **Cleanliness:** *Drips should be left to evaporate to avoid ruining the plastic surface, rather than wiped away.*
- **Alternatives:** F*or DIY tasks or attaching, products like Bostik Paneltack or other specialized acrylic-based, or acrylic nail glue may be used.*

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Type: Epoxy Resin

Epoxy resin is a high-performance, ***two-part thermosetting polymer (resin and hardener)*** that cures into a durable, rigid, and waterproof solid. It is widely used for coating, casting, adhesive, and art applications due to its high gloss, adhesion to multiple materials, and chemical resistance. It is prized for, low shrinkage, and suitability for applications like tabletop coatings and deep-pour castings.

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Type: Epoxy Resin Adhesive

Epoxy resin adhesive is a high-strength, ***two-part (resin and hardener) adhesive*** used for, filling gaps, and bonding materials like metal, wood, plastics, and ceramics. It cures by chemical reaction to form a durable, waterproof, and chemical-resistant bond that can be sanded or painted. Common types include fast-setting syringes (like Araldite, Gorilla 2-Part Epoxy) and putty for high-strength repairs.

Video: Using Epoxy Resin Adhesive

https://www.youtube.com/watch?v=oMyx4PqBLJQ

Type: Epoxy Resin Adhesive

Key Features and Applications:

- **High Bonding Strength:** Ideal for heavy-duty, structural repairs, including automotive and marine applications.
- **Versatility:** Bonds wood, metal, glass, ceramic, and most plastics.
- **Gap Filling:** Thicker formulations (gels or putties) are excellent for filling voids on vertical surfaces.
- **Durability:** Resistant to water, impacts, and extreme temperatures.
- **Application:** Surfaces must be clean, dry, and rough for the best adhesion.
- **Cure Time:** Varies from fast-setting (5–10 minutes) to full cure in 24 hours or up to 7 days for maximum strength, depending on the product.

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Type: Epoxy Resin Adhesive

Popular brands found in the UK include **Gorilla, Araldite, Everbuild, and Evo-Stik.**















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Type: Polyvinyl Acetate (PVA) Adhesive

Polyvinyl Acetate (PVA) glue is a popular, water-based, non-toxic, and versatile white or yellow adhesive commonly used for wood, paper, and porous materials. Known for its fast drying time and strong, flexible, clear-drying bond, it is favoured in woodworking and crafts. It dries in 24 hours, should not freeze, and comes in D2 (general), D3 (water-resistant), and D4 (waterproof) grades.

Video: Using Polyvinyl Acetate (PVA) Adhesive

https://www.youtube.com/watch?v=0kYWvAOyNzA

Type: Polyvinyl Acetate (PVA) Adhesive

Key Characteristics of PVA Glue:

- **Best for Porous Materials:** Ideal for wood, paper, fabric, and cardboard.
- **Woodworking Standard:** Often referred to as carpenter's glue or white glue, it creates bonds often stronger than the wood itself.
- **Types & Water Resistance:**
  - *D2: Interior use, general crafting.*
  - *D3: Water-resistant, suitable for kitchens/bathrooms.*
  - *D4: Waterproof; suitable for exterior use.*
- **Appearance:** Generally milky white, but dries to a clear or translucent finish.
- **Ease of Use:** Cleans up with water while wet.

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Type: Polyvinyl Acetate (PVA) Adhesive

Common Applications:

- **Woodworking:** Joinery, furniture repair, and laminating.
- **Bookbinding & Crafts:** Suitable for papercrafts, hobbies, and repairs.
- **DIY & Building:** Used as a sealant, primer for porous surfaces, or an additive for plaster.

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Type: Polyvinyl Acetate (PVA) Adhesive

Important Usage Tips:

- **Clamping:** PVA works best when pieces are clamped tight to ensure a thin, even layer of glue and to remove air.
- **Temperature Sensitivity:** It should not be allowed to freeze as this ruins the polymer.
- **Limitations:** It can be broken down by certain microbes, and while water-resistant variants exist, it is not ideal for submersion in water.

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Type: Hot Melt Glue

Hot melt (high-temp) glue operates around **190°C**, offering strong, durable bonds on materials like wood and metal, while **Low Melt glue** operates around **120°C**, providing safer usage and protecting delicate materials (foam, fabric) from damage. High-temp provides better penetration and strength; low-temp is best for crafting and minimizing burn risks.

We use **Low Melt Glue** in school because it is safer. The A Level coarse specification states **'Hot Melt Glue'**.

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Type: Hot Melt Glue

Hot melt adhesives (HMAs) are thermoplastic, solvent-free, solid-at-room-temperature materials that melt when heated for application, solidifying instantly upon cooling to form strong bonds. They are ideal for high-speed industrial assembly, packaging, and crafting, effectively bonding substrates like plastics, wood, cardboard, and textiles.

Video: Using Hot Melt Glue

https://www.youtube.com/watch?v=9-Gsp82aUjU

Type: Hot Melt Glue

Key Characteristics and Benefits:

- **Fast Setting:** Rapid cooling allows for quick adhesion, often in seconds, which is crucial for high-speed production lines.
- **Versatility:** Capable of bonding a wide variety of materials, including wood, paper, metal, and plastic.
- 100% Solids/Solvent-Free: Environmentally friendly with low VOC emissions, reducing the need for drying equipment.
- **Storage and Durability:** Generally have a long shelf life and provide durable bonds, with some formulations offering high heat resistance.

Type: Hot Melt Glue

Common Applications:

- **Packaging:** Case and carton sealing, labeling, and tray forming.
- **Product Assembly:** Electronics, furniture, and appliance assembly.
- **Woodworking:** Edgebanding and profiling.
- Crafting: Often used in glue guns for fast bonding.

Type: Hot Melt Glue

Common Types and Forms:

- **EVA (Ethylene Vinyl Acetate):** The most common general-purpose adhesive.
- **Polyolefins:** Known for good adhesion to plastics.
- **PUR (Polyurethane Reactive):** Offers high strength and heat resistance, curing further with moisture.
- **Forms:** Available as glue sticks, slugs, pellets, and blocks.

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Type: Hot Melt Glue

Application and Safety:

- **Hot melt adhesives** are applied using specialized equipment like glue guns or industrial bulk systems (e.g., Bostik systems). Safety precautions are necessary, as the molten glue can cause severe burns; it is crucial to use proper protective gear and avoid contact with the heated adhesive.





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Type: Cyanoacrylate (superglue)

Cyanoacrylate, commonly known as **super glue** or CA glue, is a fast-acting adhesive that creates high-strength bonds by reacting with moisture in the air or on surfaces. It is widely used for household repairs, industrial assembly, medical wound closure, and forensic fingerprinting.

Video: Using Cyanoacrylate (superglue)

https://www.youtube.com/watch?v=-4hp9BvM_AI

Type: Cyanoacrylate (superglue)

Key Characteristics:

- **Fast Curing:** Bonds typically form in seconds (ranging from 3 to 60 seconds depending on viscosity), reaching light operation strength almost instantly.
- **High Bond Strength:** The polymerisation process creates rigid chains that can withstand significant stress, often surpassing the strength of the bonded material itself.
- **Moisture-Activated:** Unlike traditional glues that dry by evaporation, cyanoacrylate hardens upon contact with water or humidity.
- **Versatility:** It bonds a wide variety of materials, including wood, metal, plastic, rubber, ceramic, and even human skin.

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Type: Cyanoacrylate (superglue)

Common Product Types and Viscosities:

The "thickness" of the glue determines its best use case:

- **Thin / Ultra-Thin:** Ideal for smooth, close-fitting surfaces and wicking into hairline cracks through capillary action.
- **Medium:** The standard general-purpose viscosity for smooth or slightly porous surfaces.
- **Thick / Gel:** Best for vertical surfaces (to prevent running), gap-filling, and highly porous materials like timber or fabrics.



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Type: Polystyrene Cement

Polystyrene cement is a specialised solvent-based adhesive designed for bonding polystyrene plastic, commonly used in hobby modeling. It works by melting the plastic surfaces, creating a ***fused***, ***"welded" joint*** rather than a ***surface bond***. Popular brands include Humbrol, Revell, and The Army Painter, available in tubes or as thin liquid applicators.







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Type: Polystyrene Cement

Key Characteristics & Usage:

- **Action:** Melts plastic surfaces to weld parts together, creating high-strength joints.
- **Compatibility:** Suitable for plastic model kits, polystyrene, and ABS plastic.
- **Types:** Available as thick glue (tubes) or thin liquid (applicator bottles) for precise application.
- **Drying Time:** Surface dries in 10-20 minutes, fully cures in 1 hour.
- **Safety:** Should be used in well-ventilated areas due to vapours.

Type: Using Polystyrene Cement

https://www.youtube.com/watch?v=I607KqYGktc

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